SPM · Term Test

CSE 421 | Session 2020-21 | Term test

Model study answers. Every listed option is addressed. Question titles are paraphrased; exact wording and original diagrams are in the source pages appended to this pack.

This is a question-setting exercise, not a conventional fixed question paper. All seven sample questions below are answered for study. The printed question-values and answer-award values are inconsistent in Task 2; the original scan is retained.

Quick revision

20 minutes; 20 marks as printed.

Task 1: create 3 questions (4 marks each), answer any 1. Printed award: 6+4 marks.

Task 2: create 2 questions (4 marks each), answer any 1. Printed award: 4+2 marks.

Task 3: create 2 questions (2 marks each), answer both. Printed award: 2+2 marks.

Questions must not be identical to or closely resemble the final-exam questions.

Instructions | Original term-test tasks and marking

Time: 20 minutes. Total: 20 marks. Task 1 asks you to create three 4-mark questions about CPI, SPI and frameworks, then answer any one; awards are shown as 3*2=6 for creation and 4 for the answer. Task 2 asks you to create two 4-mark questions about planning, scope and WBS, then answer one; awards shown are 2*2=4 and 2 for the answer, despite the earlier 4-mark wording. Task 3 asks for two 2-mark questions about COCOMO's concept and importance, then answers to both; awards shown are 2*1=2 and 2*1=2. The note disallows questions identical to or closely resembling the final exam. The seven examples here use different tasks and contexts, but the instructor decides whether they are sufficiently distinct.

Revision answer: Prepare original questions; the examples below answer every generated question, not just the required selections.

1.1 | Sample 4-mark question: assess whether a review pilot worked

Question: A team pilots peer review on one module and sees fewer escaped defects. What evidence is needed before extending the change across the organization? Explain how to avoid a misleading improvement claim.

Answer: Define the baseline, measurement period and escaped-defect rate relative to comparable size or usage. Check whether the module, team experience, release size or test effort changed at the same time. Record review effort and cycle time as well as defects, so improvement in one metric does not hide excessive cost. Replicate or extend the pilot cautiously, collect team feedback and decide whether to standardize or revise the change. Use a repeatable improvement cycle and preserve the evidence.

Revision answer: Compare normalized baseline and pilot outcomes, inspect confounds and repeat before standardizing.

1.2 | Sample 4-mark question: choose process measures

Question: A company reports 100% compliance with its new development checklist, yet customer defects keep rising. Propose four measures that could reveal whether its process improvement is useful.

Answer: Track escaped defects per release or unit of delivered functionality; track lead time from approved work to release; track change failure/recovery outcomes; and collect user-reported task problems or satisfaction for the changed product. Define each consistently and compare over a suitable period. Inspect changes in release complexity and reporting behavior. Checklist compliance is an activity measure, while these outcomes test whether the process achieves its goals.

Revision answer: Measure quality, delivery flow, release reliability and user outcomes, not compliance alone.

1.3 | Sample 4-mark question: align improvement frameworks

Question: Two teams use different improvement frameworks and want to share lessons without forcing identical workflows. Design a lightweight coordination approach.

Answer: Agree common business outcomes and a small measurement vocabulary while allowing each team to tailor its workflow. Share an improvement backlog, pilot hypotheses, baseline/results and lessons in periodic reviews. Map comparable practices across frameworks rather than equating certification labels. Adopt only changes supported by evidence and revisit them when context changes. This combines shared learning with local adaptation.

Revision answer: Share outcomes, evidence and learning while tailoring team practices.

2.1 | Sample 4-mark question: repair overlapping work packages

Question: An online booking project assigns both the payment team and notification team responsibility for sending payment receipts. How would you repair the WBS and acceptance criteria?

Answer: Identify one accountable work-package owner for the receipt capability and document the interface: payment publishes a confirmed-payment event; notification consumes it and sends the receipt. Define behavior for duplicates, failures and retries, with explicit tests and ownership. Remove duplicate scope/cost entries, ensure every necessary activity still belongs somewhere and update the WBS dictionary and estimates. Dependency timing belongs in the schedule.

Revision answer: Clarify boundaries, ownership and interfaces; remove double-counted scope.

2.2 | Sample 4-mark question: evaluate a late accessibility request

Question: A library-booking project receives a request for screen-reader support shortly before release. What planning evidence should be examined before agreeing a revised delivery commitment?

Answer: Determine whether accessibility was already a requirement or acceptance obligation; an omitted requirement is not automatically optional new scope. Identify affected pages, interaction and testing work, dependencies, specialist support and user impact. Compare delivery options, estimates, risk and resources, including phased delivery only where acceptable. Record the agreed baseline/change decision and communicate revised acceptance criteria and dates to stakeholders.

Revision answer: Check existing obligations, affected work, evidence-based estimates and an explicit change decision.

3.1 | Sample 2-mark question: compare COCOMO size revisions

Question: With unchanged Basic COCOMO mode coefficients, an estimate grows from 20 KLOC to 30 KLOC. Express the effort ratio without choosing numerical coefficients.

Answer: E1=a*20^b and E2=a*30^b. Divide to cancel a: E2/E1=(30/20)^b=1.5^b. Thus the effect depends on the size exponent; it is not necessarily a 50% effort increase.

Revision answer: Effort ratio = 1.5^b.

3.2 | Sample 2-mark question: sensitivity and estimation confidence

Question: Why should a manager present a COCOMO estimate as a range when requirements and size are uncertain? State two reasons.

Answer: First, effort depends on uncertain size, so plausible size bounds produce different effort outputs and can be amplified by the exponent. Second, calibration assumptions about the project/team may be imperfect, so one precise-looking output can hide model error. Present scenarios and update them as evidence improves.

Revision answer: Size uncertainty and model/calibration uncertainty justify a range.

Reference reading

Bob Hughes and Mike Cotterell, Software Project Management.

Roger S. Pressman and Bruce R. Maxim, Software Engineering: A Practitioner's Approach.

Barry W. Boehm, Software Engineering Economics.

Introduction - https://mhrumi.github.io/Software-project-management/01-introduction.html

Project evaluation - https://mhrumi.github.io/Software-project-management/02-project-evaluation.html

Step Wise planning - https://mhrumi.github.io/Software-project-management/03-project-planning.html

Project approach - https://mhrumi.github.io/Software-project-management/04-project-approach.html

Estimation concepts - https://mhrumi.github.io/Software-project-management/05a-estimation-concepts.html

Estimation models - https://mhrumi.github.io/Software-project-management/05b-estimation-models.html

Activity planning - https://mhrumi.github.io/Software-project-management/06-activity-planning.html

Risk management - https://mhrumi.github.io/Software-project-management/07-risk-management.html

Resource allocation - https://mhrumi.github.io/Software-project-management/08-resource-allocation.html

Monitoring and control - https://mhrumi.github.io/Software-project-management/09-monitoring-control.html

Contracts - https://mhrumi.github.io/Software-project-management/10-managing-contracts.html

Managing people - https://mhrumi.github.io/Software-project-management/11-managing-people.html

Teams and leadership - https://mhrumi.github.io/Software-project-management/12-working-in-teams.html

Software quality - https://mhrumi.github.io/Software-project-management/13-software-quality.html

These are supporting reading references, not claims that this paper or its answer key appears in those books. Course-specific terminology and source ambiguities are noted beside the affected answers.

Time: 20 Minutes Mark:20

1. Create 3 questions (4 marks each) based on the concepts of CPI & SPI and the frameworks used in this process. (3 × 2 = 6 marks). Then, answer any 1 questions. (4 marks)
2. Create 2 questions (4 marks each) from Project Planning, Scope, and Work Breakdown Structure (WBS). (2 × 2 = 4 marks). Then, answer any 1 question. (2 marks)

3. Create 2 questions (2 marks each) from the concept and importance of the COCOMO model. (2 × 1 = 2 marks). Then, answer both questions. (2 × 1 = 2 marks)

Note:

None of the questions should be identical or closely resemble the questions from your final exam paper. If any question is found to match, it will not be awarded any marks.

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